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School of Computer Science and Engineering

BCSE309L – Cryptography and Network Security

Module ~ II – Asymmetric Encryption Algorithm and Key Exchange

Elliptic Curve Cryptography

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Elliptic Curve Cryptography

- Earlier, we had a symmetric key cryptography, where only one key called private key is used.
- The sender and receiver exchange the private key. This can be vulnerable since it can eavesdrop on the communication medium.
- The solution is to use asymmetric key where two keys are used, namely, private and public keys.
- The private key is like a password and is kept secretly with each party.
- On the other hand, the public key is like an email id and is announced to the external world.
- Public key cryptography was developed by Diffie and Hellman in 1976.

- Public key cryptography is also called asymmetric key cryptography because it uses two keys:
  - private and public.
- RSA and ECC are examples of public-key cryptography.
- RSA algorithm was developed by MIT researchers Ron Rivest, Adi Shamir, and Leonard Adleman in 1977.
- RSA algorithm addresses the issue of secrecy or private key cryptography, however, it has higher computational complexity

- Elliptic Curve Cryptography (ECC) was developed by Neal Koblitz and Victor Miller who had worked at IBM in 1985.
- If the RSA algorithm uses 3072 bits public key, then the ECC algorithm uses only 256 bits public key for a comparable level of security.
- ECC algorithm uses a smaller key size and hence suitable for IoT networks. Notably, the elliptic curve has nothing todo with an ellipse. The elliptic curve is described using a quadratic curve.
- ECC is fundamentally more difficult to explain than either RSA or Diffie-Hellman.
- In ECC, addition does not mean the simple algebraic addition. Similarly, the doubling of points does not mean the simple multiplication of each coordinate by two. Nevertheless, we use the same terminologies.

ECC over Real Numbers

Elliptic curve over real number is described using

$$y^2 = x^3 + ax + b \tag{8.1}$$

where x, y, a, $b \in \mathbb{R}$, a set of real numbers and $4a^3 + 27b^2 \neq 0$. This cubic form is of degree 3.

- **ECC uses trapdoor or one way function.**
  - That is, given the public key, it is difficult to derive the private key.
- Public and private keys are generated using points of an elliptic curve.
- The curve is parametrized by {a, b, pr, nr, g}.
- Note that, a and b are the parameters of the elliptic curve.

- Here, pr imposes the maximum limits along the x and y axes.
- pr is a prime number and used in the elliptic curve over a finite field.
- nr is a cyclic group and this many times operations are repeated to get a private key.
- g is the generator or base point on the elliptic curve.
- This is used to compute other points on the curve.

Real Elliptic Curves

- ECC makes use of Elliptic curves
- Elliptic Curve is defined by mathematical functions – Cubic functions.

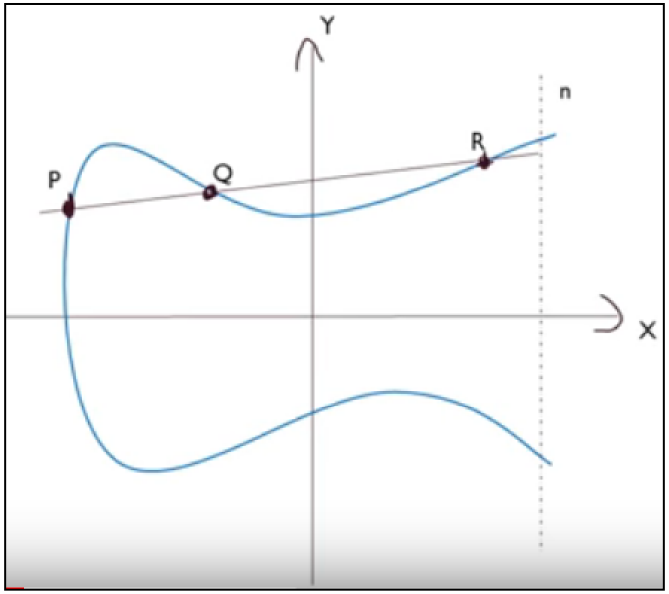
  Eg: $y^2 = x^3 + ax + b$ *[Cubic Equation] ----- (1)*

  where x, y, a, b are all real numbers

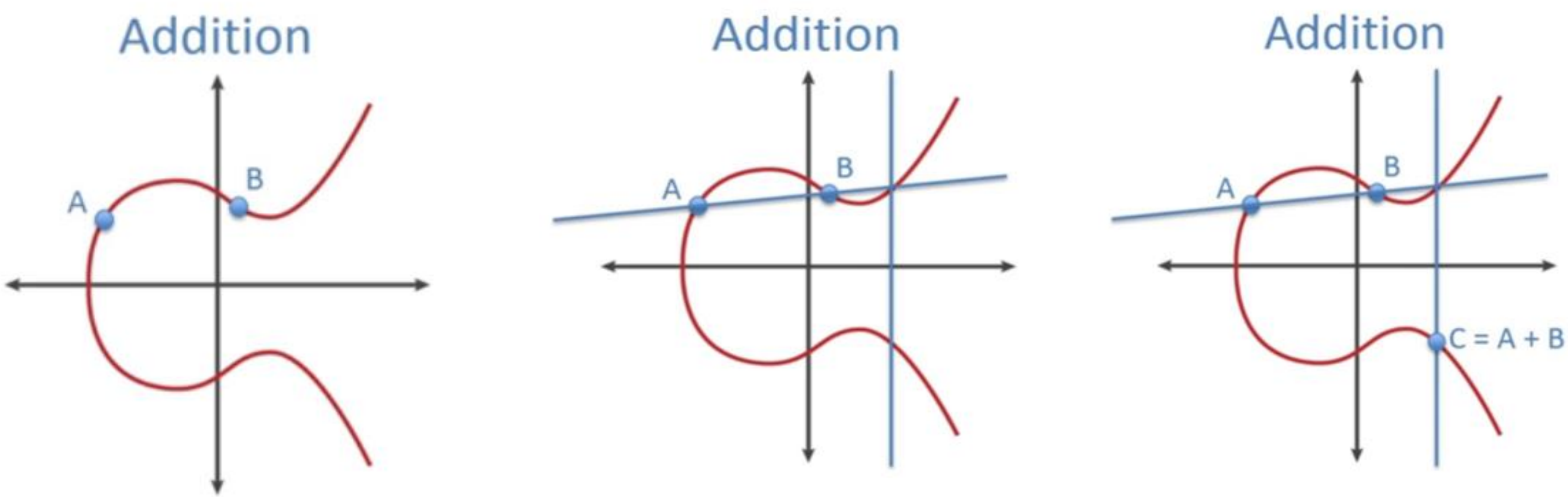
  - Symmetric to x-axis and line will intersect at 3 pts (max.)

ECC is represented by $E_p(a,b)$.

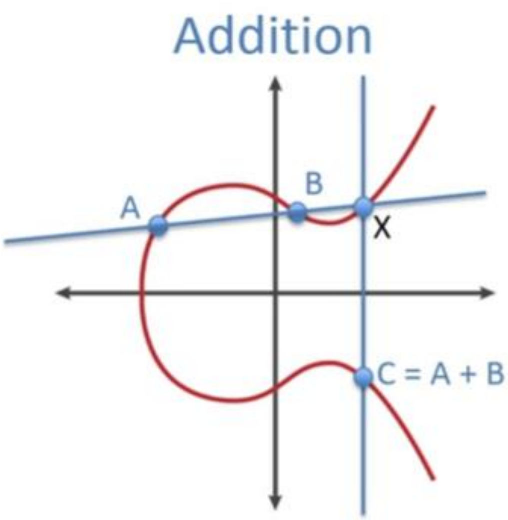
'p' is prime number and a, b are restricted to (mod p)



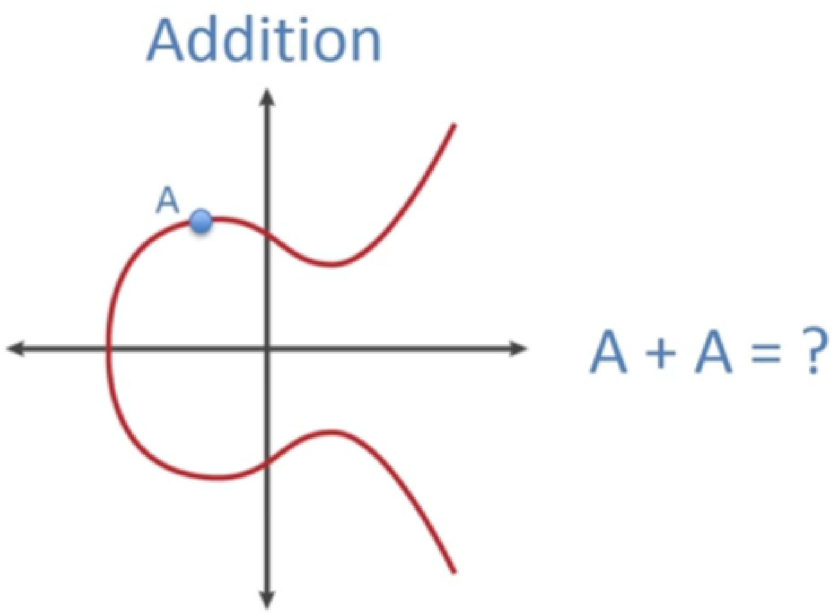
ECC – Addition (1)

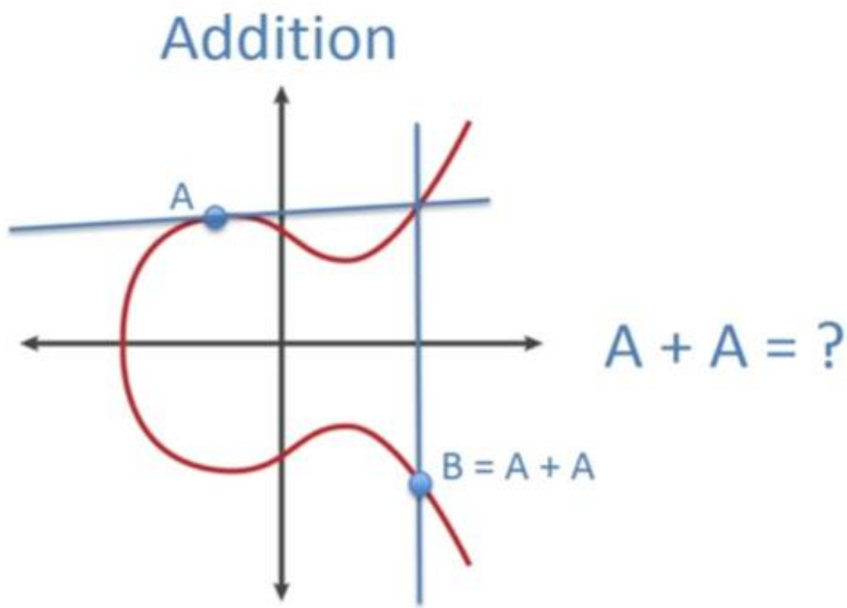


- Consider set of points E(a,b) that satisfy eqn. (1)
- Have addition operation for elliptic curve
  - Geometrically sum of A+B is reflection of the intersection X



ECC – Addition (2)

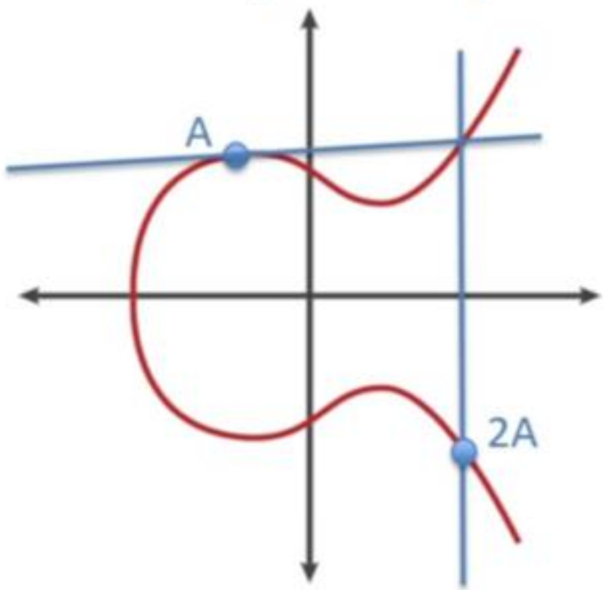




“POINT DOUBLING”

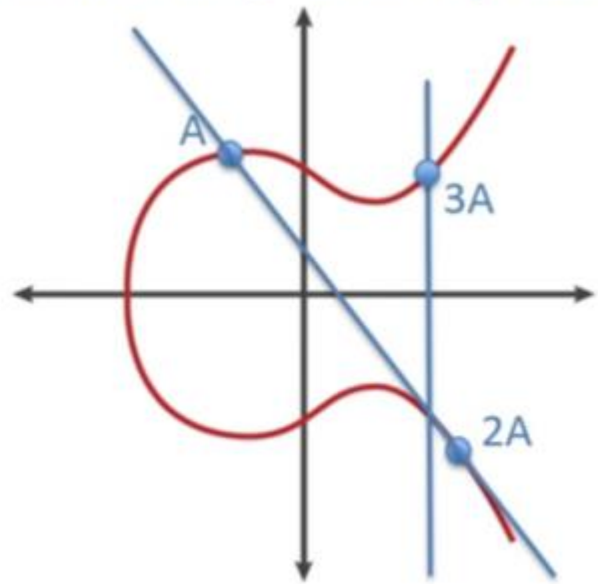
ECC – Addition (3)

Addition / Multiplication



"Point Doubling"

Addition / Multiplication



3A?

"Point Doubling"

Elliptic Curve Cryptosystem (ECC)

- Find an point in Elliptic Curve $E_{11}(1,1)$

  EC is represented as $E_P(a,b)$, so, p=11; a=1; b=1

  Elliptic curve equation is $y^2=x^3+ax+b$

  Substitute p,a,b values in this equation

  $y^2=x^3+1.\ x+1$

  $=x^3+x+1$

  x values=0, y values:+1, -1

  Points are (0,1) and (0,-1)

  Since (0,-1) is negative, take modulo p

  One of the points are (0,1), (0,10)

Elliptic Curve Cryptosystem (ECC): Addition

- Adding 2 points $A(x_a,y_a)$ and $B(x_b,y_b)$ gives $C(x_c,y_c)$

  $y^2=x^3+ax+b$

  – Find the slope (λ)

  $$\lambda=\frac{(y_b-y_a)}{(x_b-x_a)} \quad \text{if } A \neq B$$

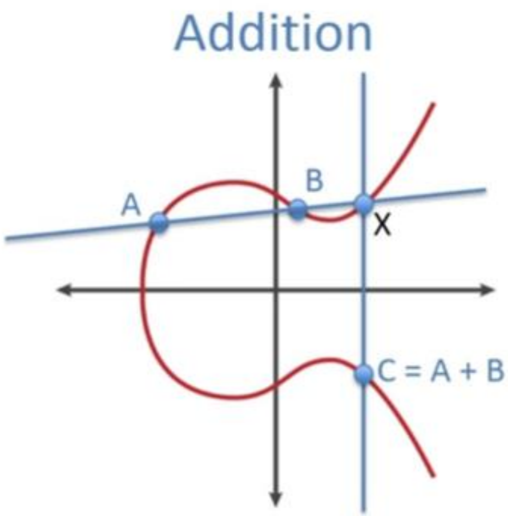
  $$\lambda=\frac{(3x_a^2+a)}{(2y_a)} \quad \text{if } A = B$$

  Where ‘a’ is obtained from $E_p(a,b)$

  – Find the sum C= A+B

  $x_c = \lambda^2 - x_a - x_b$

  $y_c = \lambda(x_a - x_c) - y_a$



Elliptic Curve Cryptosystem (ECC): Subtraction

- Negating a point

  If $B=(x_b ,y_b)$ then

  $-B=-(x_b ,y_b)=(x_b ,-y_b)$

- Subtraction
  - (A-B) can be re-written as (A+(-B))

    $(A\text{-}B) = (x_a ,y_a)-(x_b ,y_b) = (x_a ,y_a)+(x_b ,-y_b) \pmod p)$
  - Perform addition

Elliptic Curve Cryptosystem (ECC): Multiplication

- Only scalar multiplication is possible
- Multiplication between two points is not possible but repeated addition is possible
- Example

  2B=B+B

  3B=B+B+B

Note: calculate slope using $\lambda = \frac{(3x_a^2 + a)}{(2y_a)}$ if x= y

Elliptic Curve Cryptosystem (ECC): Division

- Only scalar division is possible

$$\frac{1}{a(x_a - y_a)} = a^{-1}(x_a - y_a)$$

- Multiplications steps could be followed

Elliptic Curve Cryptosystem (ECC)

- Consider two points A(3,10) and B(9,7) for $E_{23}(1,1)$ then compute
  i. A+B
  ii. 2A
  iii. 2B

Elliptic Curve Cryptosystem (ECC): Slope

- Consider two points A(3,10) and B(9,7) for $E_{23}(1,1)$ then compute

$x_a=3; y_a=10; x_b=9; y_b=7; p=23; a=1; b=1$

Find the slope (λ)

$\lambda = \frac{(yb-ya)}{(xb-xa)} \pmod p$ if A≠ B

$\frac{(7-10)}{(9-3)} = \frac{-3}{6} = \frac{-1}{2}$

a	b	q	r	s1	s2	s3	t1	t2	t3
23	2	11	1	1	0	1	0	1	-11
2	1	2	0	0	1	-2	1	-11	23

$\frac{-1}{2} \bmod 23 = -2^{-1} \pmod{23}$

$(-1)(2^{-1}) \pmod{23}$

$(-1)(-11) \pmod{23} = 11 \pmod{23} = 11$

Elliptic Curve Cryptosystem (ECC): Compute x_c

- Consider two points A(3,10) and B(9,7) for $E_{23}(1,1)$ then compute x_c

$x_a=3$; $y_a=10$; $x_b=9$; $y_b=7$; $p=23$; $a=1$; $b=1$; $\lambda=11$

Find x_c

$x_c = \lambda^2 - x_a - x_b \pmod{p}$

$x_c = 11^2-3-9 \pmod{23}$

$x_c = 109 \pmod{23}$

$x_c=17$

Elliptic Curve Cryptosystem (ECC): Compute y_c

- Consider two points A(3,10) and B(9,7) for $E_{23}(1,1)$ then compute y_c

$x_a=3;\ y_a=10;\ x_b=9;\ y_b=7;\ p=23;\ a=1;b=1;$
$\lambda=11;\ x_c=17$

Find y_c

$y_c= \lambda(x_a-x_c)-y_a \pmod{p}$

$y_c= 11*(3-17)-10 \pmod{23}$

$y_c= -164 \pmod{23}$

$y_c=20$

$A+B = C(x_c , y_c) = (17,20)$

Elliptic Curve Cryptosystem (ECC)

- Consider two point P(3,10) for $E_{23}(1,1)$ then compute 2P

To find 2 P: 2P = P+P

$\lambda = 3x_p^2 + a / 2y_p$

$= 3(3^2) + 1 / 2(10)$

$= 28/10 \bmod 23$

$= 28 \times 10^{-1} \bmod 23$

$= 14 \times 5^{-1} \bmod 23$

$\lambda = (14)(-9) \bmod 23$

$= -126 \bmod 23$

$\boxed{\lambda = 12}$

$x_r = \lambda^2 - x_p - x_q$

$= 138 \bmod 23 = 0$

$y_r = \lambda(x_p - x_r) - y_p$

$= 12(3 - 0) - 10$

$= 26$

$\boxed{2P = (0,26)}$

	1	0	23	0	1	5
4	0	1	5	1	-4	3
1	0	-4	3	-1	5	2
1	1	5	2	-1	-9	1

Elliptic Curve Cryptosystem (ECC): Slope

- Consider two points A(3,3) for $E_7(1,0)$ then compute x_c

x_a=3; y_a=3; p=7; a=1;b=0

Find the slope (λ)

$\lambda = \frac{(3x_a^2 + a)}{(2y_a)}$ if A= B

$\frac{(7-10)}{(9-3)} = \frac{-3}{6} = \frac{-1}{2}$

a	b	q	r	s1	s2	s3	t1	t2	t3
23	2	11	1	1	0	1	0	1	-11
2	1	2	0	0	1	-2	1	-11	23

$\frac{-1}{2} \bmod 23 = -2^{-1} \pmod{23}$

$(-1)(2^{-1}) \pmod{23}$

(-1)(-11) (mod 23)=11(mod 23)=11

Elliptic Curve Cryptosystem (ECC): Compute x_c

- Consider two points A(3,3) for $E_7(1,0)$ then compute x_c

$x_a=3$; $y_a=3$; $p=7$; $a=1$; $b=0$; $\lambda=7$

Find x_c

$x_c= \lambda^2-2x_a \pmod{p}$

$x_c= 11^2-3-9 \pmod{23}$

$x_c= 109 \pmod{23}$

$x_c=1$

Elliptic Curve Cryptosystem (ECC): Compute y_c

- Consider two points A(3,10) and B(9,7) for $E_{23}(1,1)$ then compute y_c

$x_a=3$; $y_a=3$; $p=7$; $a=1$; $b=0$; $\lambda=7$; $x_c=1$

Find y_c

$y_c= \lambda(x_a-x_c)-y_a \pmod p$

$y_c= 7*(3-17)-10 \pmod{23}$

$y_c= -164 \pmod{23}$

$y_c=4$

$A+B= C(x_c , y_c) = (1,4)$

Problem (For two distinct points, P≠Q)

- For $E_{29}(16, 14)$, consider the point P = (5, 25) and Q=(12, 7). Compute the value of P+Q.
- p = 29, a = 16, b = 14
- P = (5, 25), Q = (12, 7)
- $x_1 = 5, y_1 = 25,$ $x_2 = 12, y_2 = 7$

1. $X_3 = (\lambda^2 - x_1 - x_2) \bmod p$
2. $Y_3 = (\lambda (x_1 - x_3) - y_1) \bmod p$
3. $$\lambda = \frac{y_2 - y_1}{X_2 - x_1} \bmod p$$

Calculate λ value

- $$\lambda = \frac{y_2 - y_1}{x_2 - x_1} \bmod p$$

$$\lambda = \frac{7-25}{12-5} \bmod 29$$

= -18 / 7 mod 29

= -18 (1/7) mod 29

= $-18(7^{-1})$ mod 29

= -18 x 25 mod 29

= -450 mod 29

= 29 – (450 mod 29)

= 29 – 15

$\lambda =$ **14**

7^{-1} mod 29
7 x ? mod 29 = 1
Possible values are 1 to n-1, so, 1 – 28

29/7 = 4.14,
7 x 5 mod 29 = 35 mod 29 = 6 – **wrong**

Try with multiples of 29
29 x 2 = 58
7 x ? mod 58 = 1
58 / 7 = 8.2
7 x 9 mod 58 = 63 mod 58 = 5 – **wrong**

Similarly, when we try with
29 x 3, 29 x 4, 29 x5 - wrong

Finally,
29 x 6 = 174
7 x ? mod 174 = 1
174 / 7 = 24.85
7 x 25 mod 174 = 175 mod 174 = 1-Right
7 x 25 mod 29 = 1 – Right
Answer is 25

Alternative Method to find multiplicative inverse value (EEA Method)

q	a	b	r	t1	t2	t
4	29	7	1	0	1	**- 4**
7	7	1	0			

- Since it is a negative number, we need to find positive number.
- Here, moduli is 29 and final t value also a small number than moduli.
- So, add - 4 + 29 and we get 25.
- If final t value is bigger number than moduli, we can not do this addition.

Calculate x_3 value

- **$X_3 = \lambda^2 - x_1 - x_2 \bmod p$**
- $\lambda = 14$, $x_1 = 5$, $x_2 = 12$, $p = 29$
- $X_3 = 14^2 - 5 - 12 \bmod 29$
- $X_3 = 196 - 17 \bmod 29$
- $X_3 = 179 \bmod 29$
- $X_3 = 179 \bmod 29$
- **$X_3 = 5$**

Calculate y_3 value

- **$Y_3 = \lambda (x_1 - x_3) - y_1 \bmod p$**
- $\lambda = 14$, $x_1 = 5$, $x_3 = 5$, $y_1 = 25$, $p = 29$
- $Y_3 = 14 (5 - 5) - 25 \bmod 29$
- $Y_3 = - 25 \bmod 29$
- $Y_3 = 29 - (25 \bmod 29)$
- $Y_3 = 29 - 25$
- **$Y_3 = 4$**
- **Alternate Method**
- *$Y_3 = - 25 \bmod 29$*
- *When you have negative value (But, it must be smaller than moduli. Here 25 is lesser than 29, For ex, -30 is not applicable here), So, Add -25 with mod value.*

  So, -25 + 29 = 4
- *$Y_3 = 4 \bmod 29$*
- *$Y_3 = 4$.*
- ***So, (P + Q) = (5, 4)***

Checking (x_3, y_3) lies on the curve E_{29} (16, 14)

- $Y_3^2 \bmod p = x_3^3 + ax_3 + b \pmod p$
- $Y_3=4$, $p = 29$, $x_3= 5$, $a = 16$, $b = 14$
- $4^2 \bmod 29 = 5^3 + 16 \times 5 + 14 \pmod{29}$
- 16 mod 29 = 125 + 80 + 14 (mod 29)
- 16 = 219 mod 29
- **16 = 16**
- **Hence, P + Q = (5, 4)**